

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

MATHEMATICS — SSC-I (9th Grade) — ANSWER KEY (Version 1)

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks)

1. Which of the following is an irrational number?

Answer: (C) $\sqrt{5}$ 1 mark

A number is **irrational** if it cannot be expressed as a ratio p/q of two integers. $\sqrt{5}$ is irrational because 5 is not a perfect square. $5/8$ and 0.25 are rational numbers, and $\sqrt{16} = 4$ is a rational integer.

2. The property $a(b + c) = ab + ac$ is called:

Answer: (C) Distributive 1 mark

The **distributive property of multiplication over addition** states that for any real numbers a, b, c : $a(b + c) = ab + ac$. It links the two operations of addition and multiplication.

3. The set of real numbers R is the union of:

Answer: (C) Rationals & Irrationals 1 mark

The set of real numbers R is defined as $R = Q \cup Q'$, where Q is the set of **rational numbers** and Q' is the set of **irrational numbers**. Q and Q' are disjoint: $Q \cap Q' = \emptyset$.

4. The additive inverse of -7 is:

Answer: (D) 7 1 mark

The **additive inverse** of a real number a is the number $-a$ such that $a + (-a) = 0$. So the additive inverse of -7 is $+7$, because $-7 + 7 = 0$.

5. In the expression $\sqrt[3]{25}$, the number 25 is called:

Answer: (C) Radicand 1 mark

In a radical expression $\sqrt[n]{a}$, the symbol $\sqrt{}$ is the **radical sign**, n is the **index** of the radical, and a (the number under the radical sign) is the **radicand**. Here 25 is the radicand and 3 is the index.

6. The value of a^0 (where $a \neq 0$) is:

Answer: (B) 1 1 mark

By the laws of exponents, any non-zero real number raised to the power zero equals 1, i.e., $a^0 = 1$ for all $a \neq 0$. This follows from the quotient rule: $a^n/a^n = a^{n-n} = a^0 = 1$.

7. Scientific notation of 0.000052 is:

Answer: (A) 5.2×10^{-5} 1 mark

To write 0.000052 in scientific notation $c = d \times 10^n$ (where $1 \leq d < 10$), place the decimal after the first non-zero digit: 5.2. The decimal moved **5 places to the right**, so the exponent is -5 : 5.2×10^{-5} .

8. If $\log_2 64 = y$, then y equals:

Answer: (C) 6 1 mark

By definition, $\log_b x = y$ means $b^y = x$. Here $2^y = 64$. Since $64 = 2^6$, we get $y = 6$.

9. The characteristic of $\log 456.7$ is:

Answer: (C) 2 1 mark

Writing 456.7 in scientific notation: $456.7 = 4.567 \times 10^2$. The **characteristic** (integral part of the logarithm) equals the exponent of 10, which is 2.

10. $\log_b(m \times n)$ equals:

Answer: (A) $\log m + \log n$ 1 mark

This is the **Product Law of Logarithms**: $\log_b(m \cdot n) = \log_b m + \log_b n$, provided all logarithms are defined. The logarithm of a product equals the sum of the logarithms.

11. The base of the natural logarithm is:

Answer: (D) e 1 mark

The natural logarithm (also called Napierian logarithm, introduced by **John Napier**) uses the **irrational constant** $e \approx 2.71828$ as its base. It is denoted by $\ln x$, where $\ln x = \log_e x$.

12. The value of $\log_b 1$ is:

Answer: (A) 0 1 mark

Since $b^0 = 1$ for any valid base b (with $b > 0$, $b \neq 1$), by the definition of logarithm, $\log_b 1 = 0$.

SECTION B — Short Questions ($11 \times 3 = 33$ Marks)

Q.2(i): State any five basic properties of real numbers with respect to addition. Give the general form of each. 3 marks

The five basic properties of real numbers with respect to addition are: $\frac{3}{5}$ mark each

1. **Closure Property:** For all $a, b \in \mathbb{R}$, $a + b \in \mathbb{R}$.
2. **Commutative Property:** $a + b = b + a$.
3. **Associative Property:** $(a + b) + c = a + (b + c)$.
4. **Additive Identity:** $a + 0 = 0 + a = a$. (0 is the additive identity.)
5. **Additive Inverse:** $a + (-a) = 0$. ($-a$ is the additive inverse of a .)

OR

Q.2(i) OR: Define rational and irrational numbers. Write two examples of each. 3 marks

Rational numbers (Q): Numbers that can be expressed in the form p/q , where p and q are integers and $q \neq 0$. Their decimal representation is either terminating or non-terminating recurring. $1\frac{1}{2}$ marks

Examples: $3/4$, -5 , 0.75 , $0.333\ldots$

Irrational numbers (Q'): Numbers that **cannot** be expressed in the form p/q with integers p , q and $q \neq 0$. Their decimal representation is non-terminating and non-recurring. $1\frac{1}{2}$ marks

Examples: $\sqrt{2}$, π , $\sqrt{5}$, e

Q.2(ii): Identify the property used: (a) $5 \times 7 = 7 \times 5$ (b) $3 + (4 + 2) = (3 + 4) + 2$ (c) $8 \times 1 = 8$ 3 marks

- (a) $5 \times 7 = 7 \times 5 \rightarrow$ **Commutative Property of Multiplication** ($a \times b = b \times a$) 1 mark
- (b) $3 + (4 + 2) = (3 + 4) + 2 \rightarrow$ **Associative Property of Addition** ($(a + b) + c = a + (b + c)$) 1 mark
- (c) $8 \times 1 = 8 \rightarrow$ **Multiplicative Identity** ($a \times 1 = a$; 1 is the multiplicative identity) 1 mark

OR

Q.2(ii) OR: State the Trichotomy property for real numbers and verify it for $a = -2$ and $b = 5$. 3 marks

Trichotomy Property: For any two real numbers a and b , **exactly one** of the following holds:

$1\frac{1}{2}$ marks

1. $a < b$
2. $a = b$
3. $a > b$

Verification for $a = -2$, $b = 5$: $1\frac{1}{2}$ marks

- Is $-2 < 5$? **Yes** $\checkmark$
- Is $-2 = 5$? **No**
- Is $-2 > 5$? **No**

Only one condition ($-2 < 5$) is true, so the Trichotomy property holds.

Q.2(iii): Simplify: (a) $81^{1/4}$ (b) $8^{2/3}$ (c) $16^{-1/2}$ 3 marks

(a) $81^{1/4} = (3^4)^{1/4} = 3^4 \times 1/4 = 3^1 = 3$ 1 mark

(b) $8^{2/3} = (2^3)^{2/3} = 2^3 \times 2/3 = 2^2 = 4$ 1 mark

(c) $16^{-1/2} = 1/16^{1/2} = 1/\sqrt{16} = 1/4$ 1 mark

OR

Q.2(iii) OR: Write as a single radical and simplify: (a) $\sqrt[3]{6} \cdot \sqrt[3]{5}$ (b) $\sqrt{20} \div \sqrt{5}$ 3 marks

(a) Using the product rule for radicals $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$: 1½ marks

$\sqrt[3]{6} \cdot \sqrt[3]{5} = \sqrt[3]{6 \times 5} = \sqrt[3]{30}$

(b) Using the quotient rule $\sqrt{a} \div \sqrt{b} = \sqrt{a/b}$: 1½ marks

$\sqrt{20} \div \sqrt{5} = \sqrt{20/5} = \sqrt{4} = 2$

Q.2(iv): Use laws of exponents to simplify: $(a^{-2} \cdot a^5) \div a^2$ 3 marks

Using the product rule $a^m \cdot a^n = a^{m+n}$ in the numerator: 1 mark

$a^{-2} \cdot a^5 = a^{-2+5} = a^3$

Then using the quotient rule $a^m \div a^n = a^{m-n}$: 1 mark

$a^3 \div a^2 = a^{3-2} = a^1 = a$ 1 mark

OR

Q.2(iv) OR: Write $x^{3/4}$ as a radical expression and simplify if possible. 3 marks

Using the definition of rational exponent $b^{m/n} = \sqrt[n]{b^m}$, where n is the index of the radical and m is the power to which the radical is raised: 1½ marks

For $x^{3/4}$: here m = 3, n = 4.

$x^{3/4} = \sqrt[4]{x^3}$ 1½ marks

This is already in simplest form since 3 and 4 share no common factors.

Q.2(v): Find the value of x if $\sqrt{3x + 1} = 5$. 3 marks

Given: $\sqrt{3x + 1} = 5$

Squaring both sides: 1 mark

$(\sqrt{3x + 1})^2 = 5^2$

$3x + 1 = 25$

Subtracting 1 from both sides: 1 mark

$3x = 24$

Dividing both sides by 3: 1 mark

$x = 8$

OR

Q.2(v) OR: Simplify: $(2^3 \cdot 2^4) \div 2^5$ 3 marks

Using the product rule $a^m \cdot a^n = a^{m+n}$: 1 mark

$2^3 \cdot 2^4 = 2^{3+4} = 2^7$

Using the quotient rule $a^m \div a^n = a^{m-n}$: 1 mark

$2^7 \div 2^5 = 2^{7-5} = 2^2 = 4$ 1 mark

Q.2(vi): Convert to scientific notation: (a) 7,340,000 (b) 0.000089 (c) 125.6 **3 marks**

A number is in scientific notation if it is written as $d \times 10^n$, where $1 \leq d < 10$ and $n \in \mathbb{Z}$.

(a) $7,340,000 \rightarrow$ move decimal 6 places left: 7.34×10^6 **1 mark**

(b) $0.000089 \rightarrow$ move decimal 5 places right: 8.9×10^{-5} **1 mark**

(c) $125.6 \rightarrow$ move decimal 2 places left: 1.256×10^2 **1 mark**

OR

Q.2(vi) OR: Convert to standard notation: (a) 6.7×10^4 (b) 5.43×10^{-3} (c) 2.0×10^6 **3 marks**

(a) $6.7 \times 10^4 = 6.7 \times 10000 = 67,000$ **1 mark**

(b) $5.43 \times 10^{-3} = 5.43 / 1000 = 0.00543$ **1 mark**

(c) $2.0 \times 10^6 = 2.0 \times 1,000,000 = 2,000,000$ **1 mark**

Q.2(vii): Evaluate without log tables: (a) $\log_5 125$ (b) $\log_2(1/8)$ (c) $\log_{10} 10000$ **3 marks**

Using the definition $\log_b x = y$ means $b^y = x$:

(a) $\log_5 125 = \log_5 5^3 = 3$ (since $5^3 = 125$) **1 mark**

(b) $\log_2(1/8) = \log_2 2^{-3} = -3$ (since $2^{-3} = 1/8$) **1 mark**

(c) $\log_{10} 10000 = \log_{10} 10^4 = 4$ (since $10^4 = 10000$) **1 mark**

OR

Q.2(vii) OR: Find x: (a) $\log_3 x = 4$ (b) $\log_x 16 = 2$ (c) $\log_4 64 = x$ **3 marks**

(a) $\log_3 x = 4 \Rightarrow x = 3^4 = 81$ **1 mark**

(b) $\log_x 16 = 2 \Rightarrow x^2 = 16 \Rightarrow x = 4$ (taking positive root as the base of a logarithm must be positive) **1 mark**

(c) $\log_4 64 = x \Rightarrow 4^x = 64 \Rightarrow 4^x = 4^3 \Rightarrow x = 3$ **1 mark**

Q.2(viii): State the four laws of logarithms with their general form. **3 marks**

The four laws of logarithms are: **$\frac{3}{4}$ mark each**

- Product Law:** $\log_b(m \cdot n) = \log_b m + \log_b n$
- Quotient Law:** $\log_b(m / n) = \log_b m - \log_b n$
- Power Law:** $\log_b(m^n) = n \log_b m$
- Change of Base Law:** $\log_a m = \log_b m / \log_b a$

(All logarithms must be defined: base > 0 , base $\neq 1$, argument > 0 .)

OR

Q.2(viii) OR: Differentiate between common logarithm and natural logarithm (any three differences). **3 marks**

Property	Common Logarithm	Natural Logarithm
Base	10 (a rational number)	$e \approx 2.71828$ (an irrational number)
Notation	$\log x$ or $\log_{10} x$	$\ln x$ or $\log_e x$
Developed by	Henry Briggs	John Napier

1 mark per difference

Q.2(ix): If $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find: (a) $\log 6$ (b) $\log 12$ (c) $\log 18$ 3 marks

(a) $\log 6 = \log(2 \times 3) = \log 2 + \log 3 = 0.3010 + 0.4771 = \mathbf{0.7781}$ 1 mark

(b) $\log 12 = \log(2^2 \times 3) = 2 \log 2 + \log 3 = 2(0.3010) + 0.4771 = 0.6020 + 0.4771 = \mathbf{1.0791}$ 1 mark

(c) $\log 18 = \log(2 \times 3^2) = \log 2 + 2 \log 3 = 0.3010 + 2(0.4771) = 0.3010 + 0.9542 = \mathbf{1.2552}$ 1 mark

OR

Q.2(ix) OR: Find the characteristic of: (a) $\log 67.3$ (b) $\log 0.00892$ (c) $\log 5$ 3 marks

The **characteristic** is the integral part of the logarithm, equal to the exponent of 10 when the number is written in scientific notation.

(a) $67.3 = 6.73 \times 10^1 \Rightarrow \text{Characteristic} = \mathbf{1}$ 1 mark

(b) $0.00892 = 8.92 \times 10^{-3} \Rightarrow \text{Characteristic} = \mathbf{-3}$ (written as $\bar{3}$) 1 mark

(c) $5 = 5.0 \times 10^0 \Rightarrow \text{Characteristic} = \mathbf{0}$ 1 mark

Q.2(x): Use laws of logarithms to expand: $\log(x^3 y^2 \div z)$ 3 marks

Using the quotient law $\log(m/n) = \log m - \log n$:

$\log(x^3 y^2 / z) = \log(x^3 y^2) - \log z$ 1 mark

Using the product law $\log(mn) = \log m + \log n$:

$= \log(x^3) + \log(y^2) - \log z$ 1 mark

Using the power law $\log(m^n) = n \log m$:

$= \mathbf{3 \log x + 2 \log y - \log z}$ 1 mark

OR

Q.2(x) OR: Combine into a single logarithm: $2 \log x - 3 \log y + \log z$ 3 marks

Using the power law $n \log m = \log(m^n)$ on the first two terms:

$2 \log x - 3 \log y + \log z = \log(x^2) - \log(y^3) + \log z$ 1 mark

Using the product law $\log m + \log n = \log(mn)$:

$= \log(x^2 \cdot z) - \log(y^3)$ 1 mark

Using the quotient law $\log m - \log n = \log(m/n)$:

$= \mathbf{\log(x^2 z / y^3)}$ 1 mark

Q.2(xi): If $\log 2 = 0.3010$, find the number of digits in 2^{20} . 3 marks

Using the power law of logarithms:

$$\log(2^{20}) = 20 \times \log 2 \quad \text{1 mark}$$

$$= 20 \times 0.3010$$

$$= 6.020$$

So the characteristic of $\log(2^{20})$ is **6**. 1 mark

Number of digits in a whole number = **characteristic + 1**:

$$\text{Number of digits} = 6 + 1 = 7 \quad \text{1 mark}$$

OR

Q.2(xi) OR: An earthquake measured 6.5 on the Richter scale. Another measured 7.5. How many times stronger is the second? 3 marks

The Richter scale formula is $M = \log(I/I_0)$.

Let I_1, I_2 be the intensities of the first and second earthquakes:

$$\log(I_1 / I_0) = 6.5 \quad \dots(i)$$

$$\log(I_2 / I_0) = 7.5 \quad \dots(ii) \quad \text{1 mark}$$

Subtracting (i) from (ii):

$$\log(I_2 / I_0) - \log(I_1 / I_0) = 7.5 - 6.5$$

$$\log(I_2 / I_1) = 1 \quad \text{1 mark}$$

Taking antilog:

$$I_2 / I_1 = 10^1 = 10$$

So the second earthquake is **10 times stronger** than the first. 1 mark

SECTION C — Long Questions ($4 \times 5 = 20$ Marks)

Q.3(i): Use properties of exponents to simplify. Show all steps. Assume variables are positive.
 $(a^{-1/2} \cdot b^{1/3})^6 \cdot (a^2 \cdot b^{-1})$ **5 marks**

Start with the expression:

$$(a^{-1/2} \cdot b^{1/3})^6 \cdot (a^2 \cdot b^{-1})$$

Apply the **power of a product rule** $(ab)^n = a^n b^n$ to the first bracket: **1 mark**

$$= (a^{-1/2})^6 \cdot (b^{1/3})^6 \cdot a^2 \cdot b^{-1}$$

Apply the **power of a power rule** $(a^m)^n = a^{mn}$: **1 mark**

$$= a^{(-1/2)(6)} \cdot b^{(1/3)(6)} \cdot a^2 \cdot b^{-1}$$

$$= a^{-3} \cdot b^2 \cdot a^2 \cdot b^{-1}$$

Group like bases and apply the **product rule** $a^m \cdot a^n = a^{m+n}$: **1 mark**

$$= a^{-3+2} \cdot b^{2+(-1)}$$

$$= a^{-1} \cdot b^1$$
 1 mark

Write with positive exponents using $a^{-n} = 1/a^n$:

$$= b / a$$
 1 mark

OR

Q.3(i) OR: Use laws of exponents to show that: $(x^p/x^q)^{p+q} \cdot (x^q/x^r)^{q+r} \cdot (x^r/x^p)^{r+p} = 1$ **5 marks**

Apply the **quotient rule** $a^m/a^n = a^{m-n}$ inside each bracket: **1 mark**

$$\text{LHS} = (x^{p-q})^{p+q} \cdot (x^{q-r})^{q+r} \cdot (x^{r-p})^{r+p}$$

Apply the **power of a power rule** $(a^m)^n = a^{mn}$: **1 mark**

$$= x^{(p-q)(p+q)} \cdot x^{(q-r)(q+r)} \cdot x^{(r-p)(r+p)}$$

Use the identity $(a - b)(a + b) = a^2 - b^2$: **1 mark**

$$= x^{p^2 - q^2} \cdot x^{q^2 - r^2} \cdot x^{r^2 - p^2}$$

Apply the **product rule** to combine:

$$= x^{(p^2 - q^2) + (q^2 - r^2) + (r^2 - p^2)}$$
 1 mark

$$= x^0$$

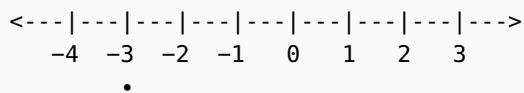
Since $a^0 = 1$ for any $a \neq 0$:

$$= 1 = \text{RHS} \quad \text{Hence proved.}$$
 1 mark

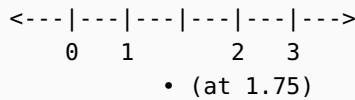
Q.3(ii): (a) Represent each on the number line: (i) -3 (ii) $7/4$ (iii) $\sqrt{5}$ (iv) $-2 < x < 3$. (b) Identify the property: (i) $(x + y) + z = x + (y + z)$ (ii) $5 \cdot (1/5) = 1$ **5 marks**

(a) Number line representations: 3 marks

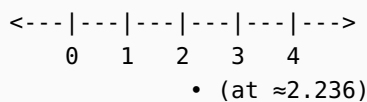
(i) -3 : A filled dot three units to the left of 0.



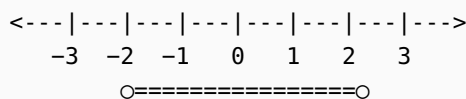
(ii) $7/4 = 1.75$: A filled dot three-quarters of the way between 1 and 2.



(iii) $\sqrt{5} \approx 2.236$: A filled dot slightly past 2, between 2 and 3.



(iv) $-2 < x < 3$: Hollow circles at -2 and 3 (exclusive), shaded line between.



(b) Identify the property: 2 marks

- (i) $(x + y) + z = x + (y + z) \rightarrow$ **Associative Property of Addition** **1 mark**
- (ii) $5 \cdot (1/5) = 1 \rightarrow$ **Multiplicative Inverse Property** ($1/5$ is the multiplicative inverse of 5) **1 mark**

OR

Q.3(ii) OR: Define: (a) closure (b) associative (c) distributive property of real numbers. Verify each using $a = 3$, $b = 4$, $c = 5$. 5 marks

(a) Closure Property: For all $a, b \in \mathbb{R}$, $a + b \in \mathbb{R}$ and $a \cdot b \in \mathbb{R}$. The sum and product of real numbers is a real number. **1 mark**

Verification with $a = 3$, $b = 4$: $3 + 4 = 7 \in \mathbb{R} \checkmark$ and $3 \times 4 = 12 \in \mathbb{R} \checkmark$ **$\frac{1}{2}$ mark**

(b) Associative Property: $(a + b) + c = a + (b + c)$ for addition, and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for multiplication. Grouping does not change the result. **1 mark**

Verification with $a = 3$, $b = 4$, $c = 5$:

- $(3 + 4) + 5 = 7 + 5 = 12$ and $3 + (4 + 5) = 3 + 9 = 12 \checkmark$
- $(3 \cdot 4) \cdot 5 = 12 \cdot 5 = 60$ and $3 \cdot (4 \cdot 5) = 3 \cdot 20 = 60 \checkmark$ **$\frac{1}{2}$ mark**

(c) Distributive Property: For all $a, b, c \in \mathbb{R}$: $a(b + c) = ab + ac$. Multiplication distributes over addition. **1 mark**

Verification with $a = 3$, $b = 4$, $c = 5$:

- LHS: $3(4 + 5) = 3 \times 9 = 27$
- RHS: $3 \times 4 + 3 \times 5 = 12 + 15 = 27$
- LHS = RHS $\checkmark$ **1 mark**

Q.3(iii): Prove the product law of logarithms: $\log_b(mn) = \log_b m + \log_b n$. Using $\log 2 = 0.3010$ and $\log 3 = 0.4771$, evaluate $\log 72$. 5 marks

Proof of Product Law: 3 marks

Let $\log_b m = x$...(i) and $\log_b n = y$...(ii)

Converting (i) and (ii) to exponential form:

$m = b^x$...(iii) and $n = b^y$...(iv) 1 mark

Multiplying (iii) and (iv):

$m \cdot n = b^x \cdot b^y = b^{x+y}$ (using product rule of exponents) 1 mark

Converting back to logarithmic form:

$\log_b(mn) = x + y$

Substituting the values of x and y from (i) and (ii):

$\log_b(mn) = \log_b m + \log_b n$ (Proved) 1 mark

Evaluate $\log 72$: 2 marks

Factorize: $72 = 8 \times 9 = 2^3 \times 3^2$

$\log 72 = \log(2^3 \times 3^2)$

$= \log(2^3) + \log(3^2)$ (product law)

$= 3 \log 2 + 2 \log 3$ (power law) 1 mark

$= 3(0.3010) + 2(0.4771)$

$= 0.9030 + 0.9542$

$= 1.8572$ 1 mark

OR

Q.3(iii) OR: Prove the quotient law of logarithms: $\log_b(m/n) = \log_b m - \log_b n$. Using $\log 2 = 0.3010$ and $\log 7 = 0.8451$, evaluate $\log(14/8)$. 5 marks

Proof of Quotient Law: 3 marks

Let $\log_b m = x$...(i) and $\log_b n = y$...(ii)

Converting to exponential form:

$m = b^x$...(iii) and $n = b^y$...(iv) 1 mark

Dividing (iii) by (iv):

$m / n = b^x / b^y = b^{x-y}$ (using quotient rule of exponents) 1 mark

Converting to logarithmic form:

$\log_b(m / n) = x - y$

Substituting values of x and y from (i) and (ii):

$\log_b(m / n) = \log_b m - \log_b n$ (Proved) 1 mark

Evaluate $\log(14/8)$: 2 marks

$\log(14/8) = \log 14 - \log 8$ (quotient law)

$= \log(2 \times 7) - \log(2^3)$

$= (\log 2 + \log 7) - 3 \log 2$ (product & power laws) 1 mark

$= \log 7 - 2 \log 2$

$= 0.8451 - 2(0.3010)$

$= 0.8451 - 0.6020$

$= 0.2431$ 1 mark

Q.3(iv): The Kansu (China) earthquake of 1920 measured 8.5 on the Richter scale. The Tokyo (Japan) earthquake of 1923 measured 7.8 on the same scale. How many times stronger was the 1920 earthquake than the 1923 earthquake? 5 marks

The Richter scale magnitude is defined as $M = \log(I / I_0)$, where I is the intensity of shock waves and I_0 is a reference constant.

Let I_1 = intensity of the 1920 Kansu earthquake and I_2 = intensity of the 1923 Tokyo earthquake.

$$M_1 = \log(I_1 / I_0) = 8.5 \quad \dots(i)$$

$$M_2 = \log(I_2 / I_0) = 7.8 \quad \dots(ii) \quad 1 \text{ mark}$$

Subtracting (ii) from (i):

$$\log(I_1 / I_0) - \log(I_2 / I_0) = 8.5 - 7.8 \quad 1 \text{ mark}$$

Using the quotient law of logarithms:

$$\log(I_1 / I_2) = 0.7 \quad 1 \text{ mark}$$

Taking antilog on both sides:

$$I_1 / I_2 = \text{antilog}(0.7) = 10^{0.7} \approx 5.01 \quad 1 \text{ mark}$$

Therefore, the 1920 Kansu earthquake was **approximately 5 times stronger** than the 1923 Tokyo earthquake. 1 mark

OR

Q.3(iv) OR: Find the number of digits in 5^{37} , given that $\log 5 = 0.6990$. 5 marks

To find the number of digits in a whole number, we use the rule:

Number of digits = characteristic of its log + 1 1 mark

Find $\log(5^{37})$ using the power law of logarithms $\log(m^n) = n \log m$:

$$\log(5^{37}) = 37 \times \log 5 \quad 1 \text{ mark}$$

$$= 37 \times 0.6990$$

$$= 25.863 \quad 1 \text{ mark}$$

The characteristic (integral part) of 25.863 is **25**. 1 mark

Therefore, the number of digits in 5^{37} :

$$= \text{characteristic} + 1 = 25 + 1 = \mathbf{26 \text{ digits}} \quad 1 \text{ mark}$$